TIDE: Trajectory-Aware Watermark Propagation for Text-to-Image Diffusion Models

Yihan Meng[1,2], **Suping Xu**[1,3], **Yanfeng Wu**[4], **Chongjun Wang**[1,2], **Lin Shang**[1,2,*]

[1]State Key Laboratory of Novel Software Technology, Nanjing University
[2]School of Computer Science, Nanjing University
[3]School of Management, Nanjing University of Posts and Telecommunications
[4]Guodian Nanjing Automation Co., Ltd.

mengyihan@smail.nju.edu.cn, suping2@ualberta.ca
yanfeng-wu@sac-china.com, {chjwang,shanglin}@nju.edu.cn

Abstract

Watermarking text-to-image diffusion models provides a practical mechanism for provenance tracking and responsible deployment. Post-processing methods are easy to deploy but vulnerable to image-space transformations. In-processing methods improve robustness by embedding the watermark within the denoising trajectory. Yet most existing methods inject the signal into the initial noise or a fixed intermediate latent, without controlling how it evolves during the remaining denoising steps. This can entangle watermark preservation with text-guided generation, risking weaker verification or degraded content fidelity. We propose TIDE, a trajectory-aware watermarking method for text-to-image diffusion models. TIDE injects a learnable watermark into an intermediate latent and jointly optimizes it with an auxiliary watermark condition, using preservation and detection objectives to limit deviation from the unwatermarked reference while maintaining separable watermark evidence. During sampling, TIDE estimates recent text-guidance directions and redirects watermark guidance toward the residual component less aligned with this local text subspace, reducing avoidable interference with semantic generation. Verification uses null-prompt partial inversion to recover the injection latent and match its masked coefficients to the target watermark. Experiments show that TIDE achieves reliable verification on unattacked images, improves robustness under common attacks, and better preserves semantic and visual fidelity.

1 Introduction

Text-to-image diffusion models [1, 2] can generate high-quality images from natural language prompts and are increasingly used for open-ended visual content creation [3]. Their deployment raises a basic provenance question. Once an image is distributed, it may be difficult to determine whether it was produced by a particular generative system and whether its source can still be verified after image attacks [4]. Diffusion watermarking methods typically embed an imperceptible signal either in the generation process [5, 6, 7, 8] or after generation [9, 10]. Verification detects whether a queried image contains the watermark through an additionally trained detector or inversion [11, 12, 7]. The central challenge is to make the watermark detectable after attacks while limiting content drift from the unwatermarked reference generation [13].

Post-processing methods embed watermarks into generated images after synthesis, either through hand-crafted frequency-domain operations [14, 15] or learned encoder-decoder architectures [16,

*Corresponding author.

40th Conference on Neural Information Processing Systems (NeurIPS 2026).

17, 18, 9]. They are model-agnostic and do not require modifying the generator. However, because the watermark is introduced after synthesis, it is not propagated by the generative trajectory. Its robustness depends largely on the trade-off between embedding strength and perceptual quality [19, 20]. In-processing methods instead embed the watermark into the diffusion sampling process, allowing the signal to evolve with the latent state [21, 6, 22, 23, 8]. This couples the watermark more tightly with the generation process, making its preservation sensitive to the denoising dynamics that govern content formation and prompt alignment.

Tree-Ring embeds a structured key into the Fourier domain of the initial diffusion noise and detects the watermark by inverting the generated image back to the noise space [6]. Since the key is introduced before denoising, it is propagated through the full sampling trajectory, which improves robustness against post-generation perturbations. However, this propagation is left to the standard denoising process, without an explicit constraint that keeps the watermarked sample close to its unwatermarked reference. In this case, the image may remain prompt-aligned while drifting from the unwatermarked generation produced from the same prompt and noise.

ROBIN mitigates this issue by moving watermark injection to an intermediate diffusion state, where coarse content has already been formed. It further optimizes a hiding prompt to keep the watermarked sample close to the reference image while preserving detectability [8, 24]. This avoids perturbing the early content-forming steps and improves content preservation compared with initial-noise watermarking [11, 12, 25]. Nevertheless, ROBIN relies on a fixed injection step and a fixed post-injection guidance rule. After injection, the remaining denoising steps are handled uniformly, although watermark residuals may interact differently with text-guidance directions at different stages [26, 27]. This limits its ability to control how the watermark signal evolves after injection.

These limitations motivate a trajectory-aware view of diffusion watermarking. We propose TIDE, a trajectory-aware watermarking method for text-to-image diffusion models that controls watermark guidance along the remaining denoising trajectory. Instead of treating watermark propagation as a passive consequence of injection, TIDE allocates watermark guidance according to its interaction with recent text-guidance directions. This preserves recoverable watermark evidence while reducing avoidable interference with prompt-driven generation.

TIDE injects a learnable watermark into selected Fourier coefficients of an intermediate latent and jointly optimizes it with an auxiliary watermark condition. The training objective combines short-term consistency, long-term preservation, and detection-oriented margin losses. The consistency and preservation losses limit deviations from the unwatermarked reference, while the margin loss maintains separable watermark evidence. This combination preserves detectability while limiting the perturbation required for watermark injection. At inference, TIDE adapts auxiliary guidance after watermark injection. At each denoising step, TIDE decomposes the watermark residual with respect to recent text-guidance directions. The component aligned with this local text subspace is treated as higher-risk for interfering with prompt-driven generation, while the complementary component is used for watermark guidance. The post-injection guidance budget is allocated according to the overlap with the text subspace, the local curvature of the text-guided trajectory, and the relative strength of the remaining watermark residual. This stepwise allocation preserves watermark evidence while limiting avoidable conflict with text guidance.

Verification follows the inversion-based setting employed by prior in-processing watermarking methods [6, 28]. Given a queried image, TIDE encodes it into latent space, partially inverts the diffusion trajectory to the injection step with an empty prompt, and compares the recovered latent with the target watermark in the masked region. The detector does not require the original prompt or the unwatermarked reference image. In the no-attack setting, TIDE reaches 1.000 AUC and 1.000 TPR@1%FPR. Under common image transformations and reconstruction attacks, it reaches 0.999 AUC and 0.986 TPR@1%FPR. It also better preserves semantic and visual fidelity compared with prior in-processing baselines. Our contributions are as follows:

- We identify post-injection trajectory interaction as a key factor in intermediate-state diffusion watermarking. Existing methods mainly specify where to inject the watermark, while TIDE studies how watermark residuals interact with text guidance during later denoising steps.
- We propose trajectory-aware watermark guidance. TIDE decomposes watermark residuals with respect to recent text-guidance directions and allocates guidance mainly to the complementary component, using an adaptive budget from local trajectory statistics.

- We introduce a learnable intermediate-latent watermark with an auxiliary watermark condition, trained by consistency, preservation, and detection-oriented margin objectives. Experiments show improved attack robustness while preserving reliable verification and visual fidelity.

2 Related work

Diffusion models. Diffusion models provide the generative backbone for modern text-to-image systems [3, 29]. Text-to-image diffusion models generate images by denoising a latent variable sampled from a Gaussian prior. A latent diffusion model samples an initial latent $z_T \sim \mathcal{N}(0, I)$ and iteratively denoises it into a clean latent z_0. The VAE decoder further maps z_0 to image space. DDPM (Denoising Diffusion Probabilistic Models) [1] formulates denoising as a probabilistic reverse process. DDIM (Denoising Diffusion Implicit Models) [2] provides a deterministic sampler whose trajectory can be approximately inverted. Prior works further show that inversion quality affects reconstruction and editing consistency [30, 31, 32]. Accurate inversion is also important for diffusion watermarking, which allows a detector to map an image back to a latent state and verify whether a hidden signal is present.

Generative watermarking. Generative watermarking embeds a verifiable signal into synthesized content for ownership verification and source attribution. Existing methods are commonly grouped by the stage at which the signal is inserted. Post-processing methods modify the final image, while in-processing methods intervene during generation.

Post-processing methods watermark the final image. Classical approaches such as DwtDct [14] and DwtDctSvd [15] modify selected frequency coefficients using discrete wavelet transform (DWT), discrete cosine transform (DCT), and singular value decomposition (SVD). Learning-based methods such as RivaGAN [33] and related encoder–decoder schemes [16, 17, 18, 9] train networks to hide and recover watermarks in images. These methods are easy to deploy because they do not require access to the generative model. However, their signal is inserted after synthesis, which exposes the watermark to image-space transformations such as compression, editing, and restoration [6, 23].

In-processing methods insert the watermark during diffusion sampling rather than modifying the final image [23, 11, 12, 6, 8, 22]. Tree-Ring [6] places a structured key in the initial noise and detects the key by inverting the generated image back to noise space. Gaussian Shading [22] also operates before denoising, but samples the initial latent through a distribution-preserving construction so that the watermarked latent follows the Gaussian prior. ROBIN [8] moves the intervention to an intermediate diffusion state and uses an optimized auxiliary prompt embedding to reduce visible artifacts while preserving detectability. Although these methods are generally more resilient than post-processing watermarks, they typically rely on passive propagation or fixed guidance after insertion. Recent studies [34, 35] also explore low-dimensional or null-space structures for diffusion watermarking, but mainly focus on the embedding subspace itself. TIDE instead focuses on post-injection trajectory control. It estimates a local text-guidance subspace during sampling and adaptively reallocates the guidance budget between semantic preservation and watermark propagation.

3 Methods

3.1 Problem setup

Our goal is to inject a watermark at an intermediate step τ while controlling its effect on the remaining denoising trajectory. Existing diffusion watermarking methods typically inject the watermark either before generation or at a predetermined denoising step. They usually propagate the perturbation with a fixed guidance scheme. This design ignores that the same perturbation can affect different parts of the denoising trajectory in different ways. Early changes may alter semantic formation, while late changes may leave few steps for the model to absorb the watermark. We treat diffusion watermarking as a trajectory-control problem rather than a one-shot latent modification.

The subsections and Figure 1 describe how we learn the watermark and the post-injection guidance mechanism, how we apply them during inference, and how we verify the watermark through inversion.

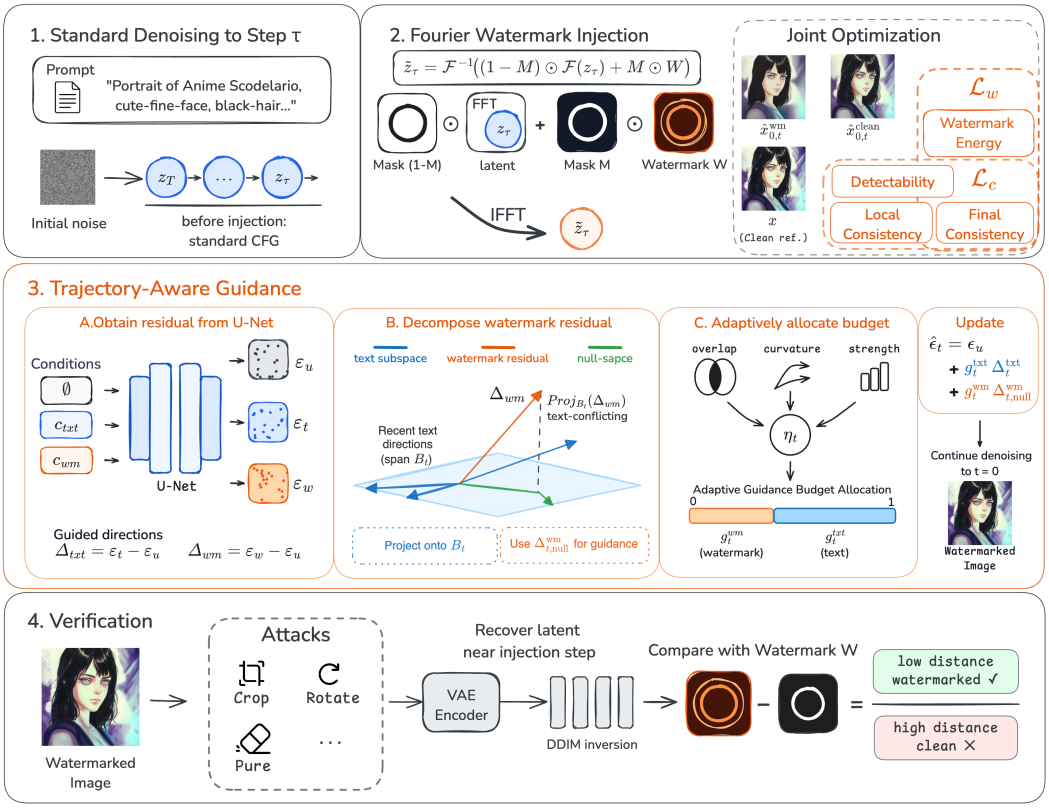


Figure 1: Overview of TIDE. A learnable watermark is injected into an intermediate latent state. After injection, TIDE projects the watermark residual away from recent text-guidance directions and assigns an adaptive guidance budget. Verification partially inverts a queried image with an empty prompt and matches the recovered Fourier coefficients to the target watermark.

3.2 Joint optimization of the watermark and watermark condition

Let $M \in \{0,1\}^d$ denote a binary mask over selected Fourier coefficients, and let W denote a learnable watermark supported on M. At the injection step τ, we overwrite the masked Fourier coefficients of the current latent:

$$\tilde{z}_\tau = \mathcal{F}^{-1}\big((1-M)\odot\mathcal{F}(z_\tau) + M\odot W\big), \tag{1}$$

where $\mathcal{F}$ and $\mathcal{F}^{-1}$ denote the Fourier transform and its inverse. Eq. (1) replaces the selected Fourier coefficients with the watermark template. We enforce Hermitian symmetry on the modified spectrum to keep $\tilde{z}_\tau$ real-valued after inversion. Sampling follows the standard text-guided trajectory before τ. At $t=\tau$, z_τ is replaced by $\tilde{z}_\tau$, and the remaining denoising steps start from $\tilde{z}_\tau$.

We jointly optimize a watermark W and a trainable auxiliary condition embedding c_{wm}. Given the injected latent $\tilde{z}_\tau$, the U-Net predicts three noise residuals: an unconditional residual $\epsilon_u = \epsilon_\theta(\tilde{z}_\tau, \tau, \emptyset)$, a watermark-conditioned residual $\epsilon_w = \epsilon_\theta(\tilde{z}_\tau, \tau, c_{\mathrm{wm}})$, and a text-conditioned residual $\epsilon_t = \epsilon_\theta(\tilde{z}_\tau, \tau, c_{\mathrm{txt}})$.

During training, we combine these three predictions with guidance scales s_{txt} and s_{wm},

$$\hat{\epsilon}^{\mathrm{wm}}_\tau = \epsilon_u + s_{\mathrm{txt}}\big(\epsilon_t - \epsilon_u\big) + s_{\mathrm{wm}}\big(\epsilon_w - \epsilon_u\big). \tag{2}$$

Substituting $\hat{\epsilon}^{\mathrm{wm}}_\tau$ into the clean-latent estimator in Section B.1 gives the one-step watermarked prediction:

$$\hat{z}^{\mathrm{wm}}_{0,\tau} = \frac{\tilde{z}_\tau - \sqrt{1-\bar{\alpha}_\tau}\,\hat{\epsilon}^{\mathrm{wm}}_\tau}{\sqrt{\bar{\alpha}_\tau}}. \tag{3}$$

For the unmodified latent z_τ, we compute the corresponding unwatermarked estimate $\hat{z}_{0,\tau}^{\text{clean}}$ using the standard text-guided prediction. Let

$$\hat{x}_{0,\tau}^{\text{wm}} = \text{Dec}(\hat{z}_{0,\tau}^{\text{wm}}) \quad \text{and} \quad \hat{x}_{0,\tau}^{\text{clean}} = \text{Dec}(\hat{z}_{0,\tau}^{\text{clean}}) \tag{4}$$

be the decoded images from the watermarked and unwatermarked one-step latents. We define an image-space reconstruction distance

$$d(x, x') = \lambda_{\text{pix}} \|x - x'\|_2^2 + \lambda_{\text{str}} \big(1 - \text{SSIM}(x, x')\big), \tag{5}$$

and use two reconstruction losses that supervise the effect of watermark injection. The short-term loss constrains the local deviation caused by injection:

$$\mathcal{L}_{\text{short}} = d\big(\hat{x}_{0,\tau}^{\text{wm}}, \hat{x}_{0,\tau}^{\text{clean}}\big)\,. \tag{6}$$

The long-term loss constrains the predicted watermarked image against the unwatermarked reference image x_{ref}:

$$\mathcal{L}_{\text{long}} = d\big(\hat{x}_{0,\tau}^{\text{wm}}, x_{\text{ref}}\big)\,. \tag{7}$$

$\mathcal{L}_{\text{short}}$ penalizes the immediate deviation introduced by watermark injection at the current step. $\mathcal{L}_{\text{long}}$ penalizes accumulated content drift relative to the unwatermarked reference image. These losses reduce visible content drift while preserving a detectable watermark signal. Let

$$D_{\text{wm}}(z; W, M) = \frac{1}{|M|} \big\| M \odot \big(\mathcal{F}(z) - W\big) \big\|_1 \tag{8}$$

denote the masked Fourier distance between a latent and the target watermark. The detection loss encourages the watermarked prediction to be closer to the target watermark than the unwatermarked prediction by a margin m:

$$\mathcal{L}_{\text{det}} = \max\Big(0,\, m + D_{\text{wm}}\big(\hat{z}_{0,\tau}^{\text{wm}}; W, M\big) - D_{\text{wm}}\big(\hat{z}_{0,\tau}^{\text{clean}}; W, M\big)\Big). \tag{9}$$

This margin separates the watermarked estimate from the unwatermarked estimate in the verification space. We also regularize the watermark energy:

$$\mathcal{L}_{\text{eng}} = \left(\frac{E(W) - E(W^{(0)})}{E(W^{(0)}) + \varepsilon}\right)^2, \qquad \text{where} \quad E(W) = \frac{1}{|M|}\|M \odot W\|_1. \tag{10}$$

This term discourages the optimized watermark from becoming excessively strong or collapsing. The watermark W and the auxiliary condition embedding c_{wm} serve different roles. We optimize them with different objectives:

$$\mathcal{L}_c = \lambda_{\text{l}} \mathcal{L}_{\text{long}} + \lambda_{\text{s}} \mathcal{L}_{\text{short}} + \lambda_d \mathcal{L}_{\text{det}}. \tag{11}$$

The energy term is applied only to W:

$$\mathcal{L}_W = \lambda_{\text{l}} \mathcal{L}_{\text{long}} + \lambda_{\text{s}} \mathcal{L}_{\text{short}} + \lambda_d \mathcal{L}_{\text{det}} + \lambda_e \mathcal{L}_{\text{eng}}. \tag{12}$$

3.3 Trajectory-aware guidance at inference

The text residual and the watermark residual change along the trajectory. The auxiliary branch may preserve the watermark at one step but interfere with the text direction at another. We design a trajectory-aware guidance scheme that reallocates the guidance budget after injection.

After injection, we obtain two conditional residuals:

$$\Delta_t^{\text{txt}} = \epsilon_t - \epsilon_u \quad \text{and} \quad \Delta_t^{\text{wm}} = \epsilon_w - \epsilon_u. \tag{13}$$

Δ_t^{txt} is the semantic update induced by the prompt, while Δ_t^{wm} is the auxiliary update that hides the watermark. Rather than applying Δ_t^{wm} directly, we separate the component that overlaps with the recent text-guidance directions. Let

$$B_t = \text{Basis}\big(\Delta_{t+1}^{\text{txt}}, \ldots, \Delta_{t+h}^{\text{txt}}\big)\,, \tag{14}$$

where h is the history size. The basis B_t represents the local semantic guidance subspace induced by recent text residuals. The watermark residual is decomposed into a projected component and a null-space component:

$$\Delta_{t,\text{proj}}^{\text{wm}} = \Pi_{B_t}(\Delta_t^{\text{wm}}) \quad \textit{and} \quad \Delta_{t,\text{null}}^{\text{wm}} = \Delta_t^{\text{wm}} - \Delta_{t,\text{proj}}^{\text{wm}}. \tag{15}$$

The projected component estimates the part of watermark guidance that is most likely to compete with text guidance. Only the null-space component is used for watermark guidance.

To allocate the post-injection guidance budget, we measure three quantities from the current trajectory. The first quantity measures semantic overlap:

$$c_t = \frac{\mathrm{RMS}\left(\Delta^{\mathrm{wm}}_{t,\mathrm{proj}}\right)}{\mathrm{RMS}(\Delta^{\mathrm{wm}}_t) + \varepsilon}. \tag{16}$$

The second measures local text-trajectory curvature:

$$\kappa_t = 1 - \cos\left(\Delta^{\mathrm{txt}}_t, \Delta^{\mathrm{txt}}_{t+1}\right). \tag{17}$$

The third measures the relative strength of the null-space watermark residual:

$$r_t = \frac{\mathrm{RMS}\left(\Delta^{\mathrm{wm}}_{t,\mathrm{null}}\right)}{\mathrm{RMS}(\Delta^{\mathrm{txt}}_t) + \varepsilon}. \tag{18}$$

$\mathrm{RMS}(\cdot)$ denotes the root-mean-square magnitude over all entries of a residual tensor. The three quantities measure semantic conflict, trajectory instability, and watermark update strength. We combine them into an adaptive factor:

$$\eta_t = (1 - c_t)\exp(-\kappa_t)\min\left(1, \frac{1}{r_t + \varepsilon}\right). \tag{19}$$

The factor η_t controls the budget assigned to watermark guidance. Let s denote the total guidance budget and let $\hat{s}$ denote the minimum budget reserved for the text branch. The watermark and text guidance weights are:

$$g^{\mathrm{wm}}_t = (s - \hat{s})\eta_t \quad \text{and} \quad g^{\mathrm{txt}}_t = s - g^{\mathrm{wm}}_t. \tag{20}$$

The post-injection noise prediction is:

$$\hat{\epsilon}_t = \epsilon_u + g^{\mathrm{txt}}_t\,\Delta^{\mathrm{txt}}_t + g^{\mathrm{wm}}_t\,\Delta^{\mathrm{wm}}_{t,\mathrm{null}}. \tag{21}$$

This update keeps the total guidance budget while reallocating it between prompt-driven generation and watermark propagation. Substituting $\hat{\epsilon}_t$ into the latent update in Section B.1 gives the next latent on the watermarked trajectory.

3.4 Watermark verification through partial inversion

Given a queried image x', verification encodes it into the latent space and inverts the trajectory to the injection step τ:

$$\hat{z}_\tau = \mathcal{I}_\tau(x'; \emptyset), \tag{22}$$

where $\mathcal{I}_\tau$ denotes DDIM inversion. Following previous methods [28, 8], we use an empty prompt during inversion with guidance scale 1.0, since the original prompt is unavailable during verification.

Verification is performed in the Fourier domain of the recovered latent. We measure the discrepancy between the recovered latent and the watermark on the masked region through Eq. (8). A smaller value of $D_{\mathrm{wm}}(\hat{z}_\tau; W, M)$ indicates stronger agreement with the implanted watermark. At query time, the detector requires the watermark parameters and the recovered latent. Unwatermarked generations are used as negative samples for threshold calibration and evaluation.

4 Experiments

4.1 Experimental setup

Models, datasets, and baselines. We use Stable Diffusion v2.1 [3] as the base text-to-image model. The watermark and the auxiliary watermark condition are optimized on prompts from the Gustavosta stable-diffusion-prompts dataset [36]. For evaluation, we use prompts from both Gustavosta and MS-COCO 2017 [37], which tests whether the learned watermark transfers across prompt distributions. We compare with post-processing methods, including DwtDct [14], DwtDctSvd [15], and RivaGAN [33], and with in-processing methods, including Stable Signature [21], Tree-Ring [6], Gaussian Shading [22], and ROBIN [8]. Tree-Ring (key) inverts only the candidate image and matches the recovered noise against the watermark key. Tree-Ring (paired) inverts both the clean and watermarked images, subtracts their recovered noises to estimate the watermark component, and matches it against the watermark regenerated from the watermark parameters.

Table 1: Main watermark verification results. We report AUC/TPR@1%FPR without attacks and mean performance over attack settings.

Category	Method	AUC/TPR@1%FPR (No attack)	AUC/TPR@1%FPR (Attacked)
Post-processing	DwtDct	0.974 / 0.624	0.574 / 0.092
	DwtDctSvd	1.000 / 1.000	0.702 / 0.262
	RivaGAN	0.999 / 0.999	0.854 / 0.448
In-processing	Tree-Ring (key)	1.000 / 1.000	0.975 / 0.694
	Tree-Ring (paired)	1.000 / 1.000	0.962 / 0.765
	Stable Signature	1.000 / 1.000	0.820 / 0.496
	Gaussian Shading	1.000 / 1.000	0.829 / 0.682
	ROBIN	1.000 / 1.000	0.981 / 0.825
	TIDE (ours)	**1.000 / 1.000**	**0.999 / 0.986**

Table 2: Image quality, perceptual distortion, prompt alignment, and distributional fidelity. FID† denotes FID between watermarked images and their corresponding unwatermarked references.

Method	PSNR↑	SSIM↑	MS-SSIM↑	CLIP-S↑	LPIPS↓	FID↓	FID† ↓
unwatermarked	∞	1.000	1.000	0.403	0	25.63	0
Tree-Ring	$15.11_{.07}$	$0.546_{.008}$	$0.589_{.005}$	$0.364_{.002}$	$0.501_{.011}$	$26.05_{.07}$	$10.84_{.03}$
ROBIN	$23.61_{.04}$	$0.757_{.010}$	$0.871_{.006}$	$0.396_{.002}$	$0.157_{.008}$	$26.86_{.09}$	$6.37_{.04}$
TIDE (ours)	$\mathbf{24.53}_{.03}$	$\mathbf{0.797}_{.011}$	$\mathbf{0.895}_{.006}$	$\mathbf{0.403}_{.004}$	$\mathbf{0.147}_{.006}$	$\mathbf{25.80}_{.07}$	$\mathbf{5.90}_{.04}$

Metrics. We report Area Under the ROC Curve (AUC) and the True Positive Rate at 1% False Positive Rate (TPR@1%FPR) for verification. Each run uses 5,000 watermarked images and 5,000 unwatermarked images. To assess the generation quality of the watermarked images, we report the Fréchet Inception Distance (FID) [38], Learned Perceptual Image Patch Similarity (LPIPS) [39], CLIP Score [40, 41], Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index (SSIM), and Multi-Scale Structural Similarity Index (MS-SSIM) [42]. The FID is calculated using 5,000 images from the MS-COCO-2017 training split.

4.2 Watermark verification and robustness

The comparative watermark verification results without attacks and under attacks are reported in Table 1. In the no-attack setting, TIDE reaches 1.000 AUC and 1.000 TPR@1%FPR, matching the strongest in-processing methods. Several methods already reach this ceiling. This setting mainly tests whether the watermark survives generation and can be recovered by inversion.

Under attacks, the methods separate more clearly. Post-processing methods degrade sharply. RivaGAN reaches 0.999 AUC and 0.999 TPR@1%FPR without attacks, but its average attacked TPR@1%FPR drops to 0.448. ROBIN is the closest method because it also uses intermediate injection and an auxiliary condition. TIDE differs in both watermark training and post-injection trajectory guidance. We jointly optimize the watermark and the auxiliary condition with preservation and detection losses. After injection, TIDE decomposes the watermark residual relative to recent text-guidance directions and adaptively assigns the watermark budget to the complementary component. Compared with ROBIN, TIDE raises the attacked average AUC from 0.981 to 0.999 and TPR@1%FPR from 0.825 to 0.986. Beyond the averaged results in Table 1, the per-attack TPR@1%FPR breakdown in Figure 7 shows where the methods separate more clearly. Tree-Ring remains strong under blur and JPEG compression but degrades under rotation and crop. ROBIN performs better on crop and reconstruction attacks, yet its TPR@1%FPR is lower under blur and rotation. TIDE maintains a high TPR@1%FPR across these settings.

Additional ablations in Appendix Table 12 and Figure 7 isolate the role of post-injection guidance. The no-guidance setting disables watermark guidance. The fixed-budget setting keeps the auxiliary branch but uses a fixed guidance budget instead of adaptive allocation. The fixed-budget setting underperforms the no-guidance setting in several cases, suggesting that fixed watermark guidance

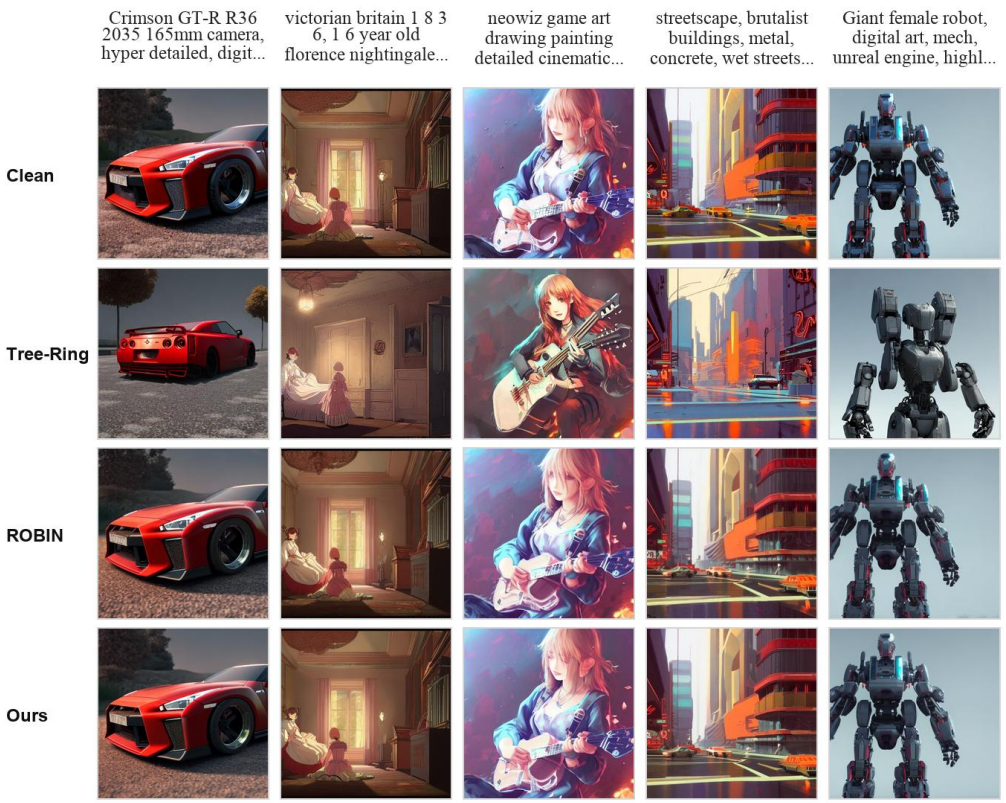


Figure 2: Qualitative comparison of different methods.

Table 3: Verification results across different injection steps. The watermark and auxiliary watermark condition are optimized at Step 35 and evaluated at different injection steps without retraining.

Steps	Blur	Noise	JPEG	Rotation	Crop	VAE-B	VAE-C	Avg
AUC								
5	0.993	0.997	0.993	**0.998**	**0.999**	0.982	0.997	0.994
15	0.996	0.998	0.997	0.997	**0.999**	0.979	**0.999**	0.995
25	0.996	0.998	0.998	**0.998**	**0.999**	0.976	**0.999**	0.995
35	**0.999**	**0.999**	**0.999**	**0.998**	**0.999**	**0.997**	**0.999**	**0.999**
45	0.885	0.995	0.997	0.899	0.991	0.976	0.996	0.963
TPR@1%FPR								
5	0.909	0.920	0.872	0.929	0.999	0.779	0.937	0.906
15	0.916	0.963	0.938	0.918	**1.000**	0.835	0.985	0.936
25	0.914	0.978	0.957	0.935	**1.000**	0.851	0.995	0.947
35	**0.983**	**0.984**	**0.983**	**0.964**	0.999	**0.991**	**0.999**	**0.986**
45	0.230	0.895	0.930	0.271	0.900	0.850	0.908	0.712

can be harmful after injection. Adaptive guidance mitigates these weaker cases by changing the watermark update according to the local text-guidance trajectory.

4.3 Image quality and content preservation

Beyond detection, we examine whether the watermark changes the generated image. We compare distortion, prompt alignment, and distributional drift against Tree-Ring and ROBIN in Table 2. TIDE achieves 24.53 PSNR, 0.797 SSIM, 0.895 MS-SSIM, and 0.147 LPIPS, consistently outperforming

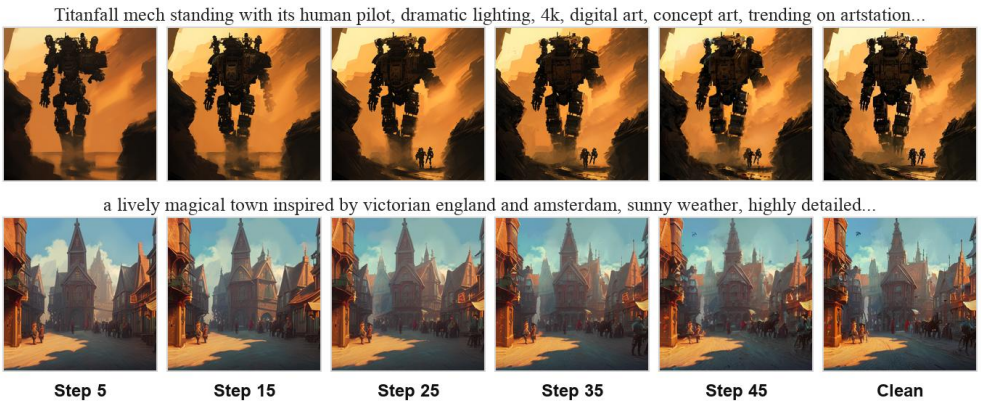


Figure 3: Visualization of cross-step transfer. The watermark and condition are trained at step 35.

competing methods on image-level metrics. It also maintains strong prompt alignment and low distributional drift, obtaining a 0.403 CLIP score, 5.90 FID†, and 25.80 FID. These results indicate that the improved robustness of TIDE is not obtained by introducing larger visible distortions.

The qualitative results show where these differences arise. Figure 2 compares unwatermarked and watermarked generations under matched prompts, with additional examples and high-frequency error maps in Figure 9 and Figure 15. Tree-Ring usually preserves the main prompt semantics, but its early watermark injection can alter object appearance, scene layout, or fine visual structure. ROBIN reduces these global changes, but the error maps still show residual differences in local texture and color. TIDE produces watermarked images that remain closer to the unwatermarked references, suggesting that it improves robustness while better preserving local visual details. The quality-robustness trade-off in Figure 8 gives a more direct view of this relation.

4.4 Cross-step transfer

We evaluate whether the watermark and its guidance condition generalize across injection steps. Both are optimized at step 35 and applied to other steps without retraining. The watermark transfers well to earlier injection steps, as shown in Table 3. Step 35 gives the best TPR@1%FPR, while other steps remain close on most attacks. The main degradation appears at step 45, especially under blur and rotation. Earlier injections leave enough denoising steps for the watermark to be incorporated into the generated image. At step 45, the image content is already largely determined, and the remaining steps mainly refine local details. This late injection makes it harder for the watermark to blend into the image, leaving it more sensitive to transformations.

Results in Figure 3 show that moving the injection later improves generation quality. Later injection perturbs fewer denoising steps and therefore better preserves the image fidelity. However, it also leaves fewer steps to propagate the watermark, which can reduce watermark performance. The quality-robustness trade-off in Figure 12 gives a direct view of this relation.

4.5 Diagnostic analysis of watermark guidance

To better understand the behavior of trajectory-aware guidance, we track the quantities that determine the post-injection watermark update over the denoising trajectory. At each denoising step, the watermark residual is decomposed relative to the recent text-guided semantic subspace. The semantic conflict score c_t quantifies the alignment between the watermark direction and the text-driven semantic direction, while the curvature term κ_t measures local instability of the semantic trajectory. The residual ratio r_t tracks the remaining strength of the semantic-null watermark component. These traces illustrate how the watermark update interacts with the text-guided denoising trajectory over time, as shown in Figure 4.

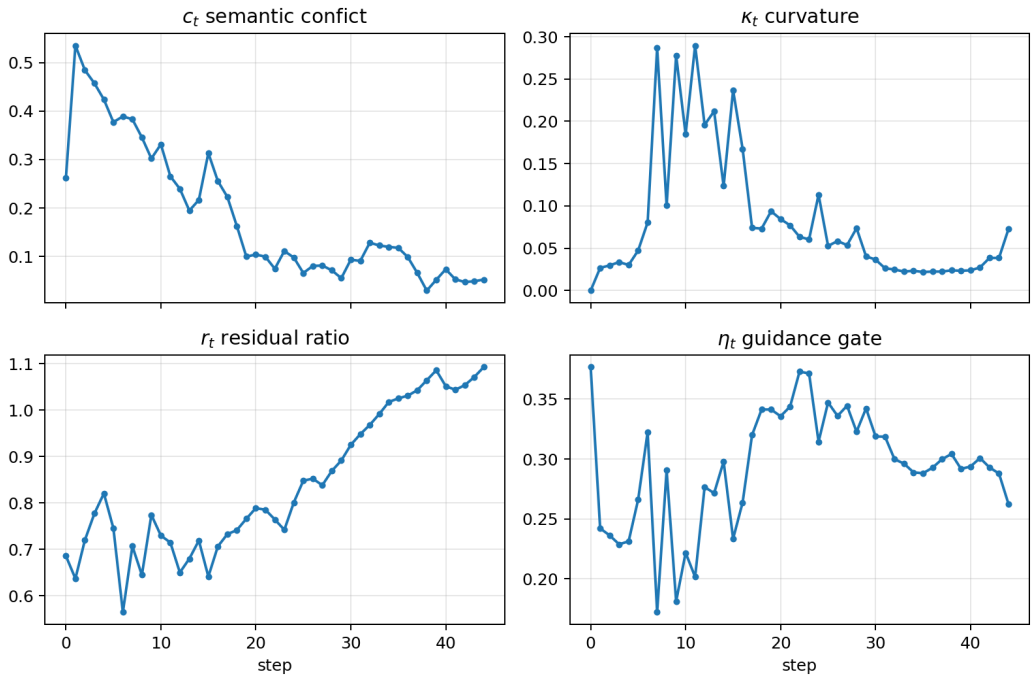


Figure 4: Diagnostic traces of the proposed watermark guidance after watermark injection. The semantic conflict score c_t, trajectory curvature κ_t, and residual ratio r_t characterize how the watermark update interacts with the text-guided denoising trajectory over time.

5 Conclusion

We introduce TIDE, a trajectory-aware watermarking method for text-to-image diffusion models. TIDE learns an intermediate watermark with an auxiliary condition and adapts post-injection denoising by allocating watermark guidance away from recent text-guidance directions. Across no-attack, attacked, and cross-step evaluations, TIDE improves verification robustness while preserving visual fidelity. These results show that robust diffusion watermarking depends not only on the injected signal but also on how the remaining denoising trajectory carries it to the final image.

6 Limitations and future work

TIDE is evaluated in the model-owner verification setting, where detection uses the generation model and a compatible inversion path. This matches prior in-processing watermarking work, but third-party verification would require a verification service or a public verifier. Although trajectory-aware guidance is compatible with multi-bit codes in principle, larger payloads may introduce stronger interference among watermark components. Another limitation is that deterministic partial inversion may be vulnerable to adaptive attacks that explicitly maximize inversion error, motivating robust or randomized verification in future work.

Acknowledgments and Disclosure of Funding

This work was supported by the National Natural Science Foundation of China (Grant No. 62606244), the Open Research Foundation of State Key Laboratory of Novel Software Technology (Grant No. KFKT2026B15), the Postgraduate Research & Practice Innovation Program of Jiangsu Province (Grant No. 26CXJH0441), and the 111 Center (Grant No. B26023).

References

[1] Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. *Advances in Neural Information Processing Systems (NeurIPS)*, 33:6840–6851, 2020.

[2] Jiaming Song, Chenlin Meng, and Stefano Ermon. Denoising diffusion implicit models. *International Conference on Learning Representations (ICLR)*, 2021.

[3] Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. High-resolution image synthesis with latent diffusion models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 10684–10695, June 2022.

[4] Jingyi Deng, Chenhao Lin, Zhengyu Zhao, Shuai Liu, Zhe Peng, Qian Wang, and Chao Shen. A survey of defenses against ai-generated visual media: Detection, disruption, and authentication. *ACM Computing Surveys*, 58(5):1–35, 2025.

[5] Jan Kociszewski, Hubert Jastrzębski, Tymoteusz Stępkowski, Filip Manijak, Krzysztof Rojek, Franziska Boenisch, and Adam Dziedzic. SERUM: Simple, efficient, robust, and unifying marking for diffusion-based image generation. In *International Conference on Learning Representations (ICLR)*, 2026.

[6] Yuxin Wen, John Kirchenbauer, Jonas Geiping, and Tom Goldstein. Tree-rings watermarks: Invisible fingerprints for diffusion images. *Advances in Neural Information Processing Systems (NeurIPS)*, 36:58047–58063, 2023.

[7] Nir Goren, Oren Katzir, Abhinav Nakarmi, Eyal Ronen, Mahmood Sharif, and Or Patashnik. Noiseprints: Distortion-free watermarks for authorship in private diffusion models. In *International Conference on Learning Representations (ICLR)*, 2026.

[8] Huayang Huang, Yu Wu, and Qian Wang. Robin: Robust and invisible watermarks for diffusion models with adversarial optimization. *Advances in Neural Information Processing Systems (NeurIPS)*, 37:3937–3963, 2024.

[9] Tu Bui, Shruti Agarwal, and John Collomosse. Trustmark: Robust watermarking and watermark removal for arbitrary resolution images. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 18629–18639, 2025.

[10] Kasra Arabi, R. Teal Witter, Chinmay Hegde, and Niv Cohen. SEAL: Semantic aware image watermarking. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 16196–16205, October 2025.

[11] Yuzhuo Chen, Zehua Ma, Han Fang, Weiming Zhang, and Nenghai Yu. Tag-wm: Tamper-aware generative image watermarking via diffusion inversion sensitivity. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 16723–16732, 2025.

[12] Hanyi Wang, Han Fang, Shi-Lin Wang, and Ee-Chien Chang. Roar: Reducing inversion error in generative image watermarking. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 19742–19751, 2025.

[13] Sihan Xu, Yidong Huang, Jiayi Pan, Ziqiao Ma, and Joyce Chai. Inversion-free image editing with language-guided diffusion models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 9452–9461, 2024.

[14] Ali Al-Haj. Combined dwt-dct digital image watermarking. *Journal of Computer Science*, 3(9): 740–746, 2007.

[15] KA Navas, Mathews Cheriyan Ajay, M Lekshmi, Tampy S Archana, and M Sasikumar. Dwt-dct-svd based watermarking. In *2008 3rd international conference on communication systems software and middleware and workshops (COMSWARE'08)*, pages 271–274. IEEE, 2008.

[16] Jiren Zhu, Russell Kaplan, Justin Johnson, and Li Fei-Fei. Hidden: Hiding data with deep networks. In *Proceedings of the European conference on computer vision (ECCV)*, pages 657–672, 2018.

[17] Matthew Tancik, Ben Mildenhall, and Ren Ng. Stegastamp: Invisible hyperlinks in physical photographs. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (CVPR)*, pages 2117–2126, 2020.

[18] Zhaoyang Jia, Han Fang, and Weiming Zhang. Mbrs: Enhancing robustness of dnn-based watermarking by mini-batch of real and simulated jpeg compression. In *Proceedings of the 29th ACM international conference on multimedia (ACM MM)*, pages 41–49, 2021.

[19] Han Fang, Kejiang Chen, Zehua Ma, Jiajun Deng, Yicong Li, Weiming Zhang, and Ee-Chien Chang. SynTag: Enhancing the geometric robustness of inversion-based generative image watermarking. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 15416–15425, October 2025.

[20] Sung Ju Lee and Nam Ik Cho. Semantic watermarking reinvented: Enhancing robustness and generation quality with fourier integrity. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 18759–18769, October 2025.

[21] Pierre Fernandez, Guillaume Couairon, Hervé Jégou, Matthijs Douze, and Teddy Furon. The stable signature: Rooting watermarks in latent diffusion models. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 22466–22477, 2023.

[22] Zijin Yang, Kai Zeng, Kejiang Chen, Han Fang, Weiming Zhang, and Nenghai Yu. Gaussian shading: Provable performance-lossless image watermarking for diffusion models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 12162–12171, 2024.

[23] Lijun Zhang, Xiao Liu, Antoni V Martin, Cindy X Bearfield, Yuriy Brun, and Hui Guan. Attack-resilient image watermarking using stable diffusion. *Advances in Neural Information Processing Systems (NeurIPS)*, 37:38480–38507, 2024.

[24] Enoal Gesny, Eva Giboulot, Teddy Furon, and Vivien Chappelier. Guidance watermarking for diffusion models. In *International Conference on Learning Representations (ICLR)*, 2026.

[25] Naresh Kumar Devulapally, Mingzhen Huang, Vishal Asnani, Shruti Agarwal, Siwei Lyu, and Vishnu Suresh Lokhande. Your text encoder can be an object-level watermarking controller. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 16576–16585, October 2025.

[26] Kushagra Pandey, Farrin Marouf Sofian, Felix Draxler, Theofanis Karaletsos, and Stephan Mandt. Variational control for guidance in diffusion models. In *Forty-second International Conference on Machine Learning (ICML)*, 2025.

[27] Tero Karras, Miika Aittala, Tuomas Kynkäänniemi, Jaakko Lehtinen, Timo Aila, and Samuli Laine. Guiding a diffusion model with a bad version of itself. *Advances in Neural Information Processing Systems (NeurIPS)*, 37:52996–53021, 2024.

[28] Ron Mokady, Amir Hertz, Kfir Aberman, Yael Pritch, and Daniel Cohen-Or. Null-text inversion for editing real images using guided diffusion models. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (CVPR)*, pages 6038–6047, 2023.

[29] Jonathan Ho and Tim Salimans. Classifier-free diffusion guidance. *arXiv preprint arXiv:2207.12598*, 2022.

[30] Xiaoyue Duan, Shuhao Cui, Guoliang Kang, Baochang Zhang, Zhengcong Fei, Mingyuan Fan, and Junshi Huang. Tuning-free inversion-enhanced control for consistent image editing. In *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, volume 38, pages 1644–1652, 2024.

[31] Dvir Samuel, Barak Meiri, Haggai Maron, Yoad Tewel, Nir Darshan, Shai Avidan, Gal Chechik, and Rami Ben-Ari. Lightning-fast image inversion and editing for text-to-image diffusion models. In *The Thirteenth International Conference on Learning Representations (ICLR)*, 2025.

[32] Jing Zuo, Luoping Cui, Chuang Zhu, and Yonggang Qi. Precise diffusion inversion: Towards novel samples and few-step models. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 38, 2025.

[33] Kevin Alex Zhang, Lei Xu, Alfredo Cuesta-Infante, and Kalyan Veeramachaneni. Robust invisible video watermarking with attention. *arXiv preprint arXiv:1909.01285*, 2019.

[34] Wenda Li, Huijie Zhang, and Qing Qu. Shallow diffuse: Robust and invisible watermarking through low-dim subspaces in diffusion models. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 38, 2025.

[35] Guokai Zhang, Lanjun Wang, Yuting Su, and An-An Liu. Markplugger: Generalizable watermark framework for latent diffusion models without retraining. *IEEE Transactions on Multimedia (TMM)*, 2025.

[36] Gustavosta. Stable-diffusion-prompts. https://huggingface.co/datasets/Gustavosta/Stable-Diffusion-Prompts, 2022.

[37] Tsung-Yi Lin, Michael Maire, Serge Belongie, James Hays, Pietro Perona, Deva Ramanan, Piotr Dollár, and C Lawrence Zitnick. Microsoft coco: Common objects in context. In *European Conference on Computer Vision (ECCV)*, pages 740–755. Springer, 2014.

[38] Martin Heusel, Hubert Ramsauer, Thomas Unterthiner, Bernhard Nessler, and Sepp Hochreiter. Gans trained by a two time-scale update rule converge to a local nash equilibrium. *Advances in Neural Information Processing Systems (NeurIPS)*, 30, 2017.

[39] Richard Zhang, Phillip Isola, Alexei A Efros, Eli Shechtman, and Oliver Wang. The unreasonable effectiveness of deep features as a perceptual metric. In *Proceedings of the IEEE conference on computer vision and pattern recognition (CVPR)*, pages 586–595, 2018.

[40] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning transferable visual models from natural language supervision. In *Proceedings of the 38th International Conference on Machine Learning (ICML)*, volume 139 of *Proceedings of Machine Learning Research*, pages 8748–8763, 2021.

[41] Christoph Schuhmann, Romain Beaumont, Richard Vencu, Cade Gordon, Ross Wightman, Mehdi Cherti, Theo Coombes, Aarush Katta, Clayton Mullis, Mitchell Wortsman, et al. Laion-5b: An open large-scale dataset for training next generation image-text models. *Advances in Neural Information Processing Systems (NeurIPS)*, 35:25278–25294, 2022.

[42] Zhou Wang, Alan C Bovik, Hamid R Sheikh, and Eero P Simoncelli. Image quality assessment: from error visibility to structural similarity. *IEEE transactions on image processing (TIP)*, 13 (4):600–612, 2004.

[43] Johannes Ballé, David Minnen, Saurabh Singh, Sung Jin Hwang, and Nick Johnston. Variational image compression with a scale hyperprior. *arXiv preprint arXiv:1802.01436*, 2018.

[44] Zhengxue Cheng, Heming Sun, Masaru Takeuchi, and Jiro Katto. Learned image compression with discretized gaussian mixture likelihoods and attention modules. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (CVPR)*, pages 7939–7948, 2020.

[45] Cheng Lu, Yuhao Zhou, Fan Bao, Jianfei Chen, Chongxuan Li, and Jun Zhu. Dpm-solver: A fast ode solver for diffusion probabilistic model sampling in around 10 steps. *Advances in Neural Information Processing Systems (NeurIPS)*, 35:5775–5787, 2022.



Figure 5: Examples of image attacks used for robustness evaluation.

A Implementation details

We evaluate robustness under five image transformations and two reconstruction attacks. Examples of transformations are shown in Figure 5. The attacks include Gaussian blur (radius=4), Gaussian noise (strength=10%), JPEG compression (quality=25), random rotation (up to 75°), 75% random cropping and rescaling, VAE-B [43], and VAE-C [44]. We follow the experimental settings of previous methods. We set $(r_{\min}, r_{\max}) = (5, 30)$. For the Stable Diffusion model, we use the default guidance scale $s = 7.5$ and the second-order multi-step DPM-Solver [45] for generation. For verification, we apply deterministic DDIM partial inversion with 50 inversion steps to recover the latent at the injection step. The learning rates for the latent watermark W and the auxiliary watermark condition c_{wm} are 0.8 and 5×10^{-4}, respectively, with a total of 2,000 optimization rounds. All experiments are conducted on NVIDIA V100 GPUs with 32GB memory.

B Preliminaries

In this section, we review DDIM sampling under classifier-free guidance, since our method modifies the guided noise prediction after injection. We then introduce DDIM inversion, which is used during verification to recover the latent state near the injection step.

B.1 Diffusion generation

Let $\{z_t\}_{t=0}^{T}$ denote a latent diffusion trajectory, where z_0 is a clean latent and larger t corresponds to a noisier latent. The forward diffusion process adds Gaussian noise according to

$$q(z_t \mid z_{t-1}) = \mathcal{N}(\sqrt{\alpha_t}\, z_{t-1}, (1-\alpha_t)I)\,, \tag{23}$$

where $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{i=1}^{t} \alpha_i$. Marginalizing the intermediate states gives

$$q(z_t \mid z_0) = \mathcal{N}\big(\sqrt{\bar{\alpha}_t}\, z_0,\, (1-\bar{\alpha}_t)I\big)\,. \tag{24}$$

Text-to-image generation learns a reverse model that gradually removes noise conditioned on the text prompt c_{txt}:

$$p_\theta(z_{0:T} \mid c_{\text{txt}}) = p(z_T) \prod_{t=1}^{T} p_\theta(z_{t-1} \mid z_t, c_{\text{txt}}), \qquad p(z_T) = \mathcal{N}(0, I). \tag{25}$$

In DDIM sampling, the reverse process is deterministic. Given a noise prediction $\hat{\epsilon}_t$ at step t, the clean-latent estimate $\hat{z}_0$ is

$$\hat{z}_0 = \frac{z_t - \sqrt{1-\bar{\alpha}_t}\,\hat{\epsilon}_t}{\sqrt{\bar{\alpha}_t}}, \tag{26}$$

and the next latent is computed as

$$z_{t-1} = \sqrt{\bar{\alpha}_{t-1}}\,\hat{z}_0 + \sqrt{1-\bar{\alpha}_{t-1}}\,\hat{\epsilon}_t. \tag{27}$$

For text-guided generation, classifier-free guidance (CFG) combines an unconditional branch and a text-conditional branch:

$$\epsilon_u = \epsilon_\theta(z_t, t, \emptyset), \qquad \epsilon_t = \epsilon_\theta(z_t, t, c_{\text{txt}}), \tag{28}$$

where $\emptyset$ denotes the null condition. With text-guidance scale s, the guided noise prediction is

$$\hat{\epsilon}_t^{\text{cfg}} = \epsilon_u + s\,(\epsilon_t - \epsilon_u)\,, \tag{29}$$

Substituting $\hat{\epsilon}_t^{\text{cfg}}$ into the DDIM update gives the standard text-guided trajectory.

$$\hat{z}_0^{\text{cfg}} = \frac{z_t - \sqrt{1-\bar{\alpha}_t}\,\hat{\epsilon}_t^{\text{cfg}}}{\sqrt{\bar{\alpha}_t}}, \tag{30}$$

$$z_{t-1}^{\text{cfg}} = \sqrt{\bar{\alpha}_{t-1}}\,\hat{z}_0^{\text{cfg}} + \sqrt{1-\bar{\alpha}_{t-1}}\,\hat{\epsilon}_t^{\text{cfg}}. \tag{31}$$

Our method modifies Eq. (29) after watermark injection.

B.2 DDIM inversion

For an image x, its latent representation is obtained by the VAE encoder, $z_0 = \mathcal{E}(x)$. DDIM inversion reconstructs a noisy trajectory from z_0 by applying the deterministic update in the opposite temporal direction. At step t, given condition c, the noise prediction is written as

$$\hat{\epsilon}_t(x, c) = \epsilon_\theta(z_t, t, c). \tag{32}$$

The deterministic inversion update is

$$\hat{z}_{0,t} = \frac{z_t - \sqrt{1-\bar{\alpha}_t}\,\hat{\epsilon}_t}{\sqrt{\bar{\alpha}_t}}, \qquad \text{and} \qquad z_{t+1} = \sqrt{\bar{\alpha}_{t+1}}\,\hat{z}_{0,t} + \sqrt{1-\bar{\alpha}_{t+1}}\,\hat{\epsilon}_t. \tag{33}$$

Therefore, DDIM sampling and DDIM inversion share the same latent estimator and differ only in traversal direction. In this work, inversion is used to recover the latent state at the watermark injection step.

C Watermark construction

We describe how the latent watermark support and watermark template are constructed in Algorithm 1. Following Fourier-domain diffusion watermarks such as Tree-Ring [6], we place the watermark on an annular support rather than on arbitrary latent coefficients. Unlike initial-noise watermarking, however, our watermark is injected at the intermediate latent z_τ and is later optimized together with the auxiliary watermark condition. The annular support localizes the verification signal in the Fourier watermark domain while avoiding a spatially localized artifact in the generated image.

Let $z_\tau \in \mathbb{R}^{C\times H\times W}$ denote the latent at the injection step. We use the Fourier domain as the watermark domain and write $\mathcal{F}$ and $\mathcal{F}^{-1}$ for the Fourier transform and its inverse. The support is determined by a private channel set $\mathcal{C}$ and an annular region $\mathcal{R}$. In our implementation, $\mathcal{R}$ is centered at the Fourier-grid center and is specified by inner and outer radii $(r_{\min}, r_{\max})$. These quantities are construction parameters used by the detector, rather than public model parameters.

The binary support mask is used to select which Fourier coefficients are replaced. Hermitian symmetry is enforced on the complex watermark template and on the resulting watermarked Fourier latent, rather than by symmetrizing the binary mask itself.

D Optimization algorithm

The optimization procedure is summarized in Algorithm 2. The VAE, U-Net, and text encoder are kept frozen. The trainable variables are the latent watermark target W and, when enabled, the auxiliary watermark condition c_{wm}. Each iteration recovers the latent at the embedding point, injects the current watermark, evaluates clean and watermarked denoising branches, and updates the trainable variables using the objectives introduced in Section 3.2.

The projection step keeps the optimized watermark inside the same support and structural family used during construction.

Algorithm 1 Watermark construction

Input: latent z_τ; channel set $\mathcal{C}$; radii $(r_{\min}, r_{\max})$; initialization prior Π
Output: mask M; initialized watermark $W^{(0)}$; watermarked latent $\tilde{z}_\tau$

$o \leftarrow (\text{Height}/2, \text{Width}/2)$
$\mathcal{R} \leftarrow \{(u,v) : r_{\min} \le \|(u,v) - o\|_2 \le r_{\max}\}$
$M \leftarrow \mathbf{1}\{(c,u,v) : c \in \mathcal{C},\ (u,v) \in \mathcal{R}\}$
$x \leftarrow \text{Sample}(\Pi)$
$W^{(0)} \leftarrow \mathcal{F}(x)$
$W^{(0)} \leftarrow \text{RingProject}(W^{(0)}, \mathcal{C}, r_{\min}, r_{\max})$
$W^{(0)} \leftarrow \text{Hermitian}(W^{(0)})$
$Z_\tau \leftarrow \mathcal{F}(z_\tau)$
$\tilde{Z}_\tau \leftarrow Z_\tau$
$\tilde{Z}_\tau[M] \leftarrow W^{(0)}[M]$
$\tilde{Z}_\tau \leftarrow \text{Hermitian}(\tilde{Z}_\tau)$
$\tilde{z}_\tau \leftarrow \mathcal{F}^{-1}(\tilde{Z}_\tau)$
return $(M, W^{(0)}, \tilde{z}_\tau)$

Algorithm 2 Joint Optimization of Watermark and Watermark Condition

Input: Prompt set $\mathcal{P}$; frozen models $(\text{Dec}, \psi, \epsilon_\theta)$; frozen sampler $\mathcal{S}_\theta$; mask M; initial watermark $W^{(0)}$; initial condition $c_{\text{wm}}^{(0)}$
Output: Optimized watermark $W^\star$ and condition $c_{\text{wm}}^\star$

$k \leftarrow 0$
while not converged **do**
  $p \sim \mathcal{P}, \quad z_T \sim \mathcal{N}(0, I), \quad \tau = \tau_{train}$
  $(\{z_t^{\text{ref}}\}_{t=T}^{0}, x_{\text{ref}}) \leftarrow \mathcal{S}_\theta(z_T, p; s)$
  $z_\tau \leftarrow z_\tau^{\text{ref}}, \quad c_p \leftarrow \psi(p), \quad c_\emptyset \leftarrow \psi(\emptyset)$
  $\tilde{z}_\tau \leftarrow \text{Inject}(z_\tau, W^{(k)}, M)$
  Compute $\hat{z}_{0,\tau}^{\text{clean}}$ from z_τ using standard text-guided CFG
  Compute $\hat{z}_{0,\tau}^{\text{wm}}$ from $\tilde{z}_\tau$ using text guidance and $c_{\text{wm}}^{(k)}$
  $L_{\text{short}} \leftarrow d(\text{Dec}(\hat{z}_{0,\tau}^{\text{wm}}), \text{Dec}(\hat{z}_{0,\tau}^{\text{clean}}))$
  $L_{\text{long}} \leftarrow d(\text{Dec}(\hat{z}_{0,\tau}^{\text{wm}}), x_{\text{ref}})$
  $L_{\text{det}} \leftarrow \max\big(0, m + D_{\text{wm}}(\hat{z}_{0,\tau}^{\text{wm}}; W^{(k)}, M) - D_{\text{wm}}(\hat{z}_{0,\tau}^{\text{clean}}; W^{(k)}, M)\big)$
  $L_{\text{eng}} \leftarrow \left(\frac{E(W^{(k)}) - E(W^{(0)})}{E(W^{(0)}) + \epsilon}\right)^2$
  $L_W \leftarrow \lambda_l L_{\text{long}} + \lambda_s L_{\text{short}} + \lambda_d L_{\text{det}} + \lambda_e L_{\text{eng}}$
  $L_c \leftarrow \lambda_l L_{\text{long}} + \lambda_s L_{\text{short}} + \lambda_d L_{\text{det}}$
  $\bar{W}^{(k+1)} \leftarrow \text{OptStep}_W(W^{(k)}, \nabla_W L_W)$
  $c_{\text{wm}}^{(k+1)} \leftarrow \text{OptStep}_c(c_{\text{wm}}^{(k)}, \nabla_c L_c)$
  $W^{(k+1)} \leftarrow P_{M,\Pi}(\bar{W}^{(k+1)})$
  $k \leftarrow k + 1$
end while
return $W^{(k)}, c_{\text{wm}}^{(k)}$

D.1 Auxiliary watermark condition is not a natural-language prompt

The auxiliary watermark condition c_{wm} is optimized together with the latent watermark target W while the VAE, U-Net, and text encoder are kept frozen. Although c_{wm} is used to form an auxiliary conditional branch during denoising, it should not be interpreted as a natural-language prompt. In TIDE, the auxiliary branch produces the watermark residual

$$\Delta_t^{\text{wm}} = \epsilon_\theta(z_t, t, c_{\text{wm}}) - \epsilon_\theta(z_t, t, \emptyset), \tag{34}$$

which has the same residual form as the text-guidance direction

$$\Delta_t^{\text{txt}} = \epsilon_\theta(z_t, t, c_{\text{txt}}) - \epsilon_\theta(z_t, t, \emptyset). \tag{35}$$

Table 4: Nearest-token projection diagnostics for the optimized auxiliary watermark condition c_{wm}. The tokenizer and text encoder are matched to each Stable Diffusion backbone. The learned condition remains close to the empty-prompt condition, while nearest-token projection produces low-similarity, non-semantic token fragments.

Backbone	$\cos(c_{\mathrm{wm}}, c_{\emptyset})$	Nearest-token mean / max	Nearest-token projection example
SD1.5	0.9545	0.1514 / 0.1611	`domain niss niss parent parent ... jad...`
SD2.0	0.9135	0.1278 / 0.1452	`lect respective ukip salsa citrus ... roses...`
SD2.1	0.9337	0.1329 / 0.1503	`lect broom graded commodity citrus roses roses uil`

This makes the auxiliary branch guidance-like in its effect on the denoising trajectory. However, c_{wm} is optimized directly in the continuous conditioning space and is not constrained to lie on the discrete text-token manifold. It is therefore better understood as a learned continuous auxiliary condition for watermark propagation, rather than as a textual prompt.

To examine whether the learned condition corresponds to meaningful text, we project the optimized auxiliary condition to its nearest vocabulary tokens. For each backbone, the tokenizer and text encoder are matched to the corresponding Stable Diffusion model. We report the cosine similarity between the optimized auxiliary condition and the empty-prompt condition, together with the mean and maximum nearest-token cosine similarities. The results are shown in Table 4.

Across the three backbones, the optimized auxiliary watermark condition remains close to the empty-prompt condition, with cosine similarity above 0.91. This indicates that the learned condition behaves as a continuous perturbation around the null condition rather than moving toward a specific semantic prompt. At the same time, the nearest-token similarities are low: the mean similarity is only about 0.13–0.15, and the maximum similarity is about 0.15–0.16. Such values do not indicate a reliable projection onto discrete vocabulary tokens.

The projected token sequences are also not semantically meaningful. They contain repeated or fragmented tokens such as `niss`, `lect`, `citrus`, `roses`, and `uil`, and they do not form stable natural-language descriptions related to the watermarking objective. This suggests that the auxiliary branch is text-guidance-like in mechanism, because it contributes an additional conditional residual during denoising, but the learned auxiliary watermark condition is not textual in representation. Accordingly, we describe c_{wm} as a learned continuous auxiliary condition rather than a decodable natural-language prompt.

E Additional results

We provide additional results and visualizations to support the quantitative evaluation. These results give a more direct view of runtime, backbone transfer, noise-to-image generation, compound attacks, cross-step transfer, guidance behavior, and local high-frequency changes.

E.1 Runtime comparison

Table 5: Generation and verification time comparison.

Method	Generation Time (s/image)	Verification Time (s/query)
Stable Signature	2.2944	0.1570
Tree-Ring (key)	2.2351	2.1632
Tree-Ring (paired)	2.2351	2.9513
ROBIN	2.3173	0.7530
Ours	2.2917	0.7395

Runtime mainly differs in the verification stage, since the compared methods use different detector inputs and inversion depths. Generation time remains similar across methods, around 2.2–2.3 seconds per image, indicating that TIDE adds little overhead during sampling. For verification, the reported time follows the input protocol of each detector. Tree-Ring (key) inverts only the candidate image and

matches the recovered initial noise against the watermark key. Tree-Ring (paired) additionally uses the paired clean image to estimate the watermark component. ROBIN and TIDE invert the candidate image with an empty prompt. Unwatermarked generations are used for threshold calibration and evaluation. The main difference is the inversion depth: Tree-Ring inverts to the initial noise, whereas ROBIN and TIDE only invert to the injection step, set to step 35 in our experiments. As a result, TIDE achieves a verification time close to ROBIN and lower than Tree-Ring. Stable Signature is the fastest because it uses a learned extractor rather than diffusion inversion.

E.2 Evaluation on different diffusion backbones

We further evaluate whether TIDE remains effective across different Stable Diffusion backbones. Across SDv1.5, SDv2, and SDv2.1, the watermark is reliably detected on clean images and remains robust after attacks. The quality metrics also stay within a comparable range, showing that TIDE can be applied across Stable Diffusion versions without relying on a specific backbone. The full verification and quality results are shown in Tables 6 and 7.

Table 6: Verification performance across diffusion backbones.

Model	**AUC/TPR@1%FPR (Clean)**	**AUC/TPR@1%FPR (Attacked)**	**Generation (s/image)**	**Verification (s/query)**
SDv1.5	1.000/1.000	0.995/0.916	2.7329	0.8423
SDv2	1.000/1.000	0.994/0.941	2.3133	0.7282
SDv2.1	1.000/1.000	0.999/0.986	2.2917	0.7395

Table 7: Image quality across diffusion backbones. FID† denotes FID between watermarked images and their corresponding unwatermarked references, following the protocol in Table 2.

Model	**PSNR↑**	**SSIM↑**	**MS-SSIM↑**	**CLIP-S↑**	**LPIPS↓**	**FID† ↓**
w/o w (SDv1.5)	∞	1.000	1.000	0.362	0	0
w/o w (SDv2 & SDv2.1)	∞	1.000	1.000	0.403	0	0
SDv1.5	26.01	0.757	0.856	0.367	0.162	7.31
SDv2	23.91	0.763	0.875	0.400	0.153	6.42
SDv2.1	24.53	0.797	0.895	0.403	0.147	5.90

E.3 Noise-to-image generation

We test TIDE beyond the text-to-image setting by applying it to noise-to-image generation. This setting removes the text prompt from generation. It tests whether TIDE can preserve detectability when generation is not conditioned on a text prompt. The results show that TIDE retains strong verification performance under both image transformations and reconstruction attacks. The gain is especially clear under rotation, where TIDE substantially improves TPR@1%FPR over ROBIN while maintaining high AUC. Detailed results are provided in Tables 8 and 9.

Table 8: AUC for Noise-to-Image generation.

Method	Blur	Noise	JPEG	Rotation	Crop	VAE-B	VAE-C	Avg
ROBIN	0.996	0.997	1.000	0.992	1.000	0.991	0.999	0.996
TIDE	0.999	1.000	1.000	0.999	1.000	0.996	0.999	0.999

E.4 Multiple simultaneous attacks

Real-world manipulations may combine several distortions rather than applying a single isolated attack. We therefore evaluate a compound-attack setting where multiple random attacks are applied to the same generated image. Verification becomes more difficult as more attacks are combined,

Table 9: TPR@1%FPR for Noise-to-Image generation.

Method	Blur	Noise	JPEG	Rotation	Crop	VAE-B	VAE-C	Avg
ROBIN	1.000	1.000	1.000	0.845	1.000	0.978	0.999	0.975
TIDE	0.999	1.000	1.000	0.999	1.000	0.979	1.000	0.997

but TIDE retains stronger detectability than Tree-Ring and ROBIN across all attack counts, as summarized in Table 10.

Table 10: AUC under different numbers of simultaneous random attacks.

Method	1	2	3	4	5	6
Tree-Ring	0.969	0.809	0.699	0.520	0.546	0.509
ROBIN	0.973	0.814	0.759	0.579	0.558	0.556
Ours	0.999	0.926	0.899	0.746	0.705	0.592

E.5 Additional qualitative comparison across methods

Averaged robustness can hide which attacks separate the compared methods. We therefore provide per-attack AUC and TPR@1%FPR results for the in-processing baselines and guidance ablations in Figures 6 and 7. The quality-robustness view in Figure 8 further shows how verification strength changes with visual fidelity. Additional matched-prompt examples are provided in Figure 9.

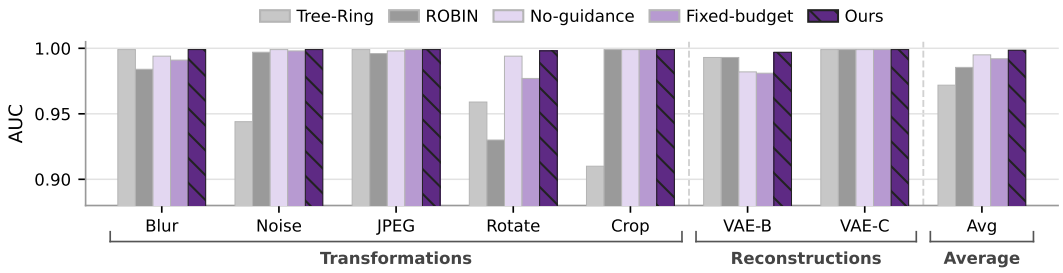


Figure 6: AUC under image transformations and reconstruction attacks. The comparison includes two in-processing methods and two ablations.

E.6 Comparison with subspace-based watermarking

Recent subspace-based watermarking methods provide an alternative design for diffusion watermarking by embedding the watermark in a low-dimensional or null-space component. This line of work is closely related to TIDE, as both aim to preserve image quality while maintaining a detectable signal in the diffusion latent space. The key difference is that subspace-based methods mainly control where the watermark is embedded, whereas TIDE further controls how the injected watermark evolves during the subsequent denoising trajectory. In this setting, Shallow Diffuse [34] serves as a representative recent method from this line of work.

The comparison in Table 13 shows that subspace-based embedding is effective under several photometric and reconstruction attacks, confirming the strength of this design. However, geometric transformations remain more challenging, where the watermark signal is more likely to be spatially misaligned after manipulation. TIDE achieves more stable AUC across the attack suite, especially under rotation and crop, suggesting that trajectory-aware post-injection guidance provides additional robustness beyond selecting the embedding subspace.

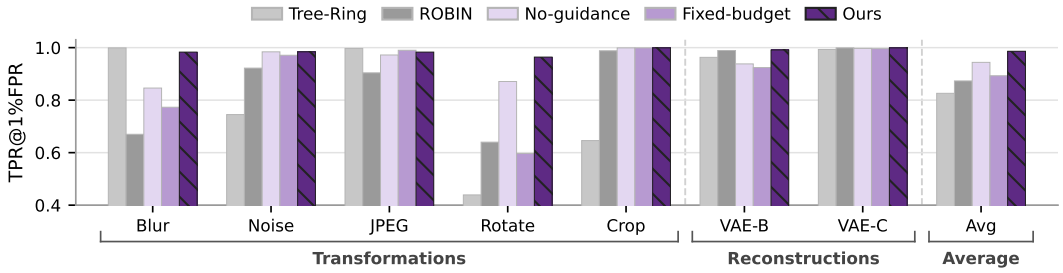


Figure 7: TPR@1%FPR under image transformations and reconstruction attacks. The comparison includes two in-processing methods and two ablations.

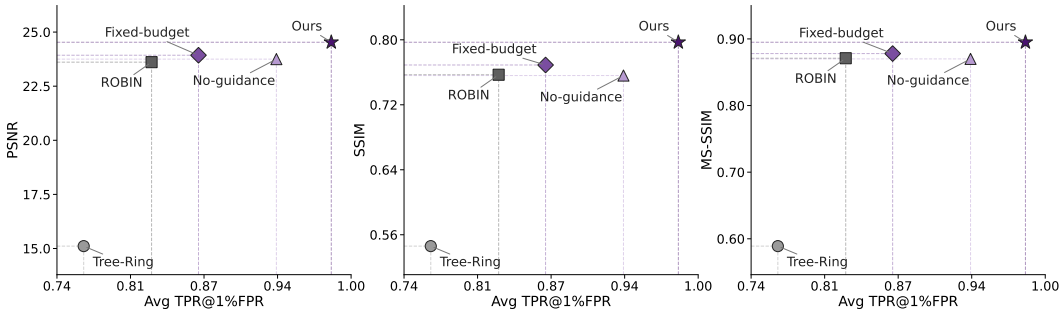


Figure 8: Quality-robustness trade-off across watermarking methods. Points closer to the upper-right indicate better image quality and stronger robustness.

Table 11: AUC for guidance ablations and cross-step transfer under image transformations.

Method	Steps	Blur	Noise	JPEG	Rotation	Crop	AVG
No-guidance	5	0.994	0.995	0.989	0.998	0.999	0.995
	15	0.995	0.996	0.995	0.998	0.999	0.997
	25	0.997	0.998	0.997	0.998	0.999	0.998
	35	0.993	0.999	0.998	0.994	0.999	0.997
	45	0.879	0.993	0.996	0.900	0.991	0.952
Fixed-budget	5	0.992	0.997	0.999	0.944	1.000	0.986
	15	0.995	0.998	0.999	0.958	1.000	0.990
	25	0.995	0.998	0.999	0.977	1.000	0.994
	35	0.991	0.999	0.999	0.977	0.999	0.993
	45	0.875	0.994	0.997	0.878	0.989	0.947
Ours	5	0.993	0.997	0.993	0.998	0.999	0.996
	15	0.996	0.998	0.997	0.997	0.999	0.997
	25	0.996	0.998	0.998	0.998	0.999	0.998
	35	0.999	0.999	0.999	0.998	0.999	0.999
	45	0.885	0.995	0.997	0.899	0.991	0.953

Table 12: TPR@1%FPR for guidance ablations and cross-step transfer under image transformations.

Method	Steps	Blur	Noise	JPEG	Rotation	Crop	AVG
No-guidance	5	0.882	0.904	0.793	0.947	0.999	0.905
	15	0.903	0.943	0.845	0.948	1.000	0.928
	25	0.909	0.970	0.938	0.966	1.000	0.957
	35	0.846	0.984	0.972	0.871	0.999	0.934
	45	0.202	0.878	0.908	0.248	0.890	0.625
Fixed-budget	5	0.860	0.935	0.993	0.336	1.000	0.825
	15	0.859	0.955	0.990	0.363	1.000	0.833
	25	0.865	0.954	0.991	0.617	1.000	0.885
	35	0.773	0.971	0.990	0.597	0.999	0.866
	45	0.193	0.897	0.928	0.186	0.853	0.611
Ours	5	0.909	0.920	0.872	0.929	0.999	0.926
	15	0.916	0.963	0.938	0.918	1.000	0.947
	25	0.914	0.978	0.957	0.935	1.000	0.957
	35	0.983	0.984	0.983	0.964	0.999	0.983
	45	0.230	0.895	0.930	0.271	0.900	0.645

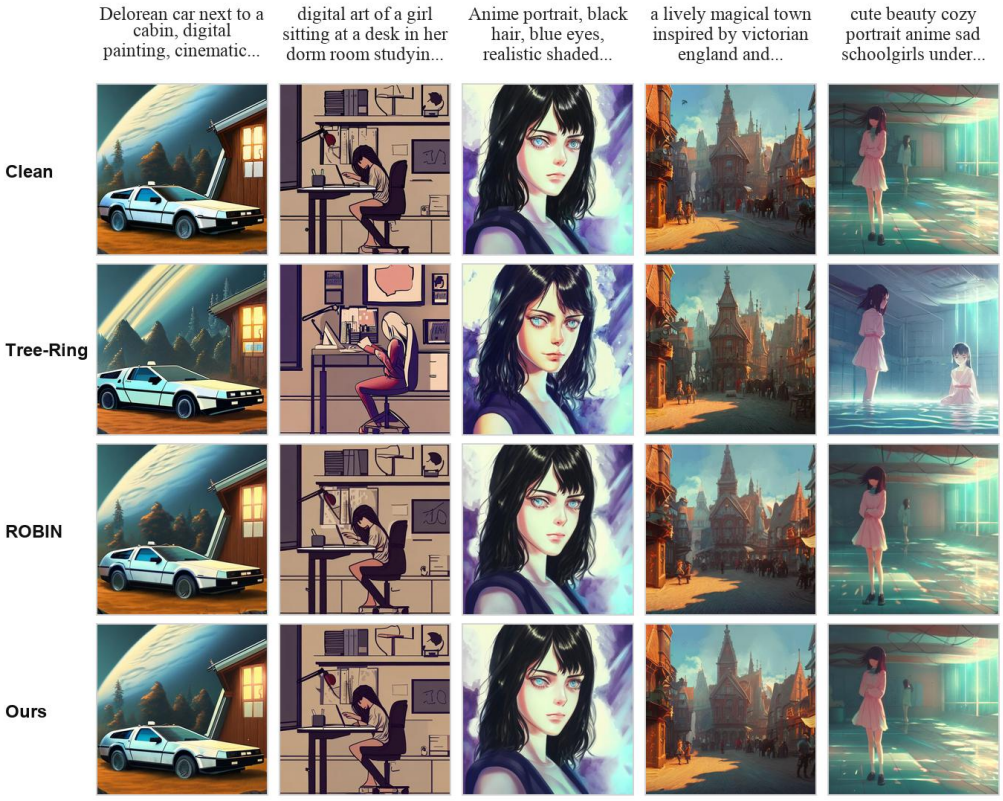


Figure 9: Additional qualitative comparison under matched prompts.

Table 13: Comparison with ROBIN and Shallow Diffuse.

Method	Blur	Noise	JPEG	Rotation	Crop	VAE-B	VAE-C	Avg
ROBIN($r \in [5, 30]$)	0.997	0.984	0.996	0.930	0.999	0.993	0.999	0.985
Shallow Diffuse($r \in [0, 10]$)	0.999	0.997	0.999	0.691	0.582	1.000	0.998	0.895
TIDE($r \in [0, 10]$)	0.994	0.996	0.992	0.994	0.999	0.995	0.999	0.996
TIDE($r \in [5, 30]$)	0.999	0.999	0.999	0.998	0.999	0.997	0.999	0.999

E.7 Additional qualitative results for cross-step transfer

Cross-step transfer measures whether the optimized watermark and auxiliary condition remain effective when the injection step changes at inference time. We provide the full AUC and TPR@1%FPR breakdowns in Figures 10 and 11. To connect robustness with image fidelity, we also report image-quality metrics in Table 14 and the corresponding quality-robustness view in Figure 12. Additional matched-prompt examples are provided in Figure 13.

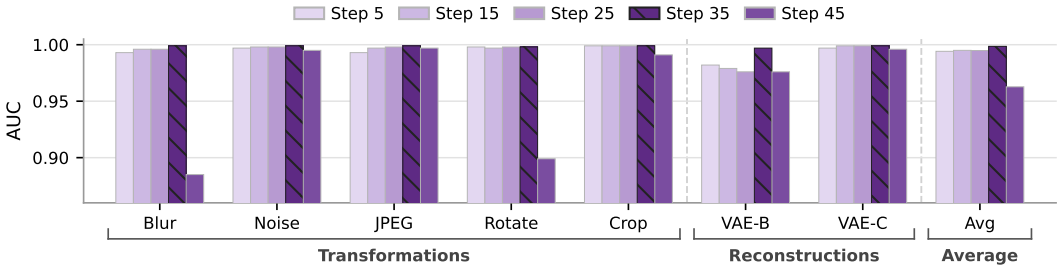


Figure 10: Cross-step transfer measured by AUC. The watermark and watermark condition are trained at Step 35 and evaluated by injecting them at other steps under attacks.

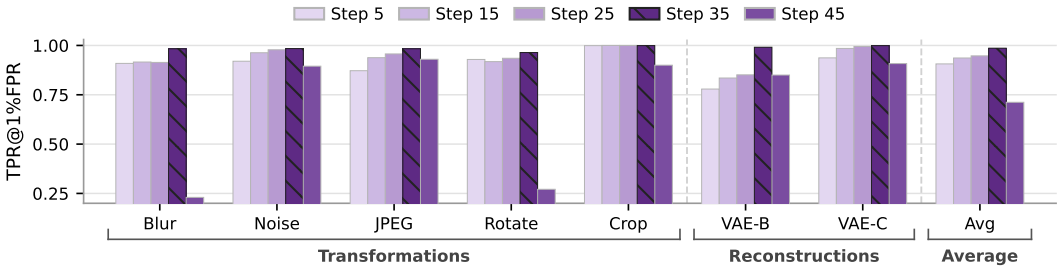


Figure 11: Cross-step transfer measured by TPR@1%FPR. The watermark and watermark condition are trained at Step 35 and evaluated by injecting them at other steps under image attacks.

Table 14: Image quality results for guidance ablations and cross-step transfer.

Method	Steps	PSNR	SSIM	MS-SSIM
No-guidance	5	17.39	0.573	0.611
	15	19.75	0.630	0.719
	25	22.02	0.691	0.808
	35	23.75	0.756	0.870
	45	23.45	0.774	0.871
Fixed-budget	5	15.46	0.510	0.572
	15	18.51	0.606	0.702
	25	21.59	0.697	0.809
	35	23.93	0.769	0.878
	45	23.61	0.779	0.875
Ours	5	18.92	0.582	0.637
	15	20.31	0.640	0.763
	25	23.19	0.702	0.827
	35	24.53	0.797	0.895
	45	24.21	0.814	0.893

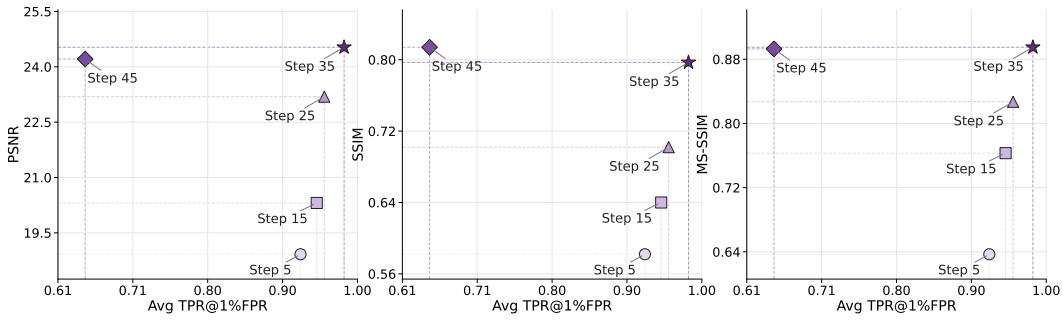


Figure 12: Quality-robustness trade-off across injection steps. Points closer to the upper-right indicate better image quality and stronger attacked verification.



Figure 13: Additional qualitative cross-step transfer results.

E.8 Qualitative results for guidance ablations

The post-injection guidance step is designed to preserve the injected watermark trajectory while reducing visible artifacts. We therefore include additional ablations that separate the effect of guidance from the choice of injection step. Guidance consistently improves the quality-robustness trade-off, and the trend remains stable under cross-step transfer. The complete per-attack AUC and TPR@1%FPR results are given in Tables 11 and 12, with additional visualizations in Figures 6, 7, and 8. The visual effect of post-injection guidance is shown in Figure 14.

E.9 High-frequency error maps

We include high-frequency error maps to make local image changes easier to inspect. We compute each high-frequency error map from an unwatermarked reference image x and the corresponding watermarked image $\tilde{x}$. Let G_σ be a Gaussian low-pass filter and $H_\sigma(x) = x - G_\sigma(x)$ be the

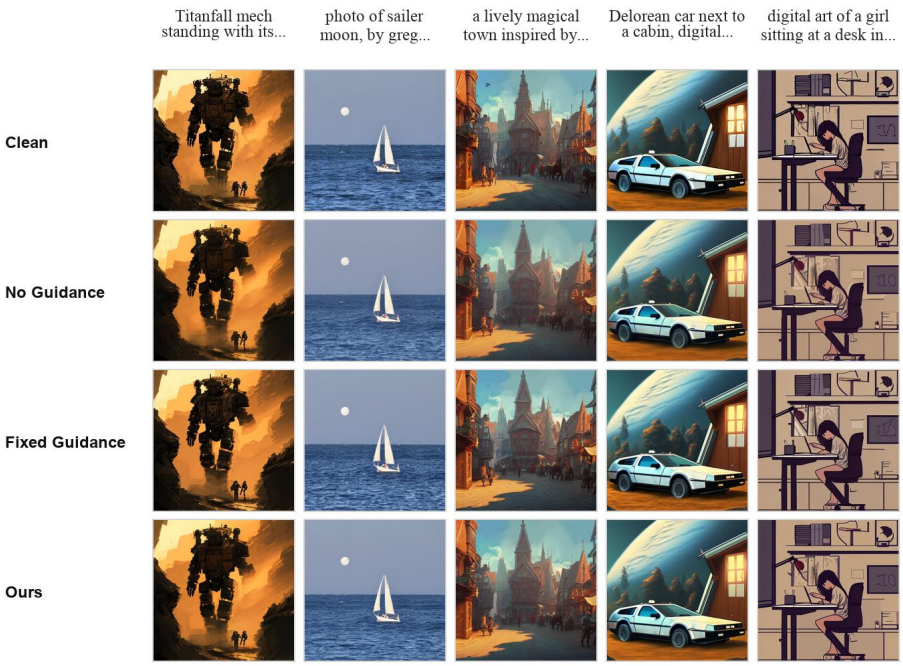


Figure 14: Qualitative effect of post-injection guidance.

corresponding high-pass residual. The visualized map is

$$E_{\mathrm{HF}}(u,v) = \frac{1}{3}\sum_{c=1}^{3} \left| H_\sigma(\tilde{x})_{u,v,c} - H_\sigma(x)_{u,v,c} \right| . \tag{36}$$

The map is percentile-clipped and normalized for display only.

Method-level results are shown in Figure 15 and Figure 16. Cross-step results are shown in Figure 17 and Figure 18. Guidance ablation results are shown in Figure 19.

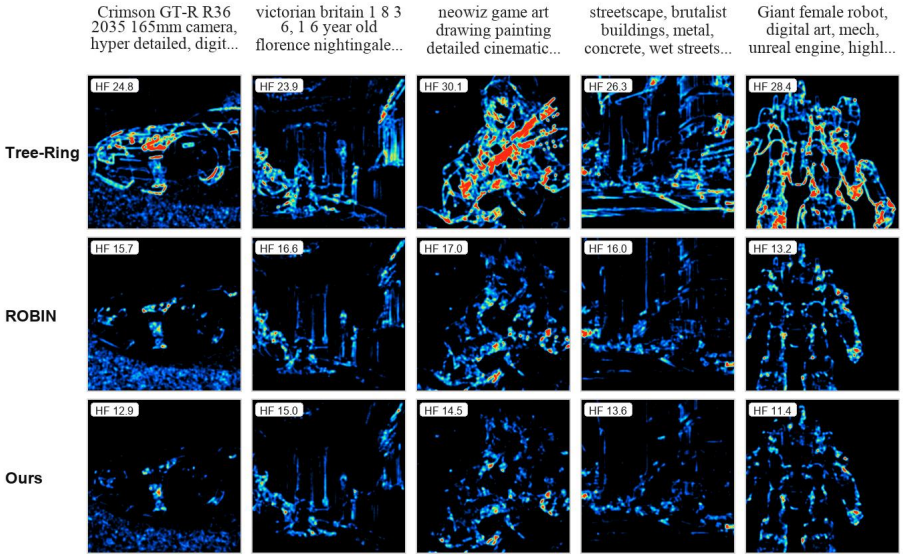


Figure 15: High-frequency error maps for watermarking methods.

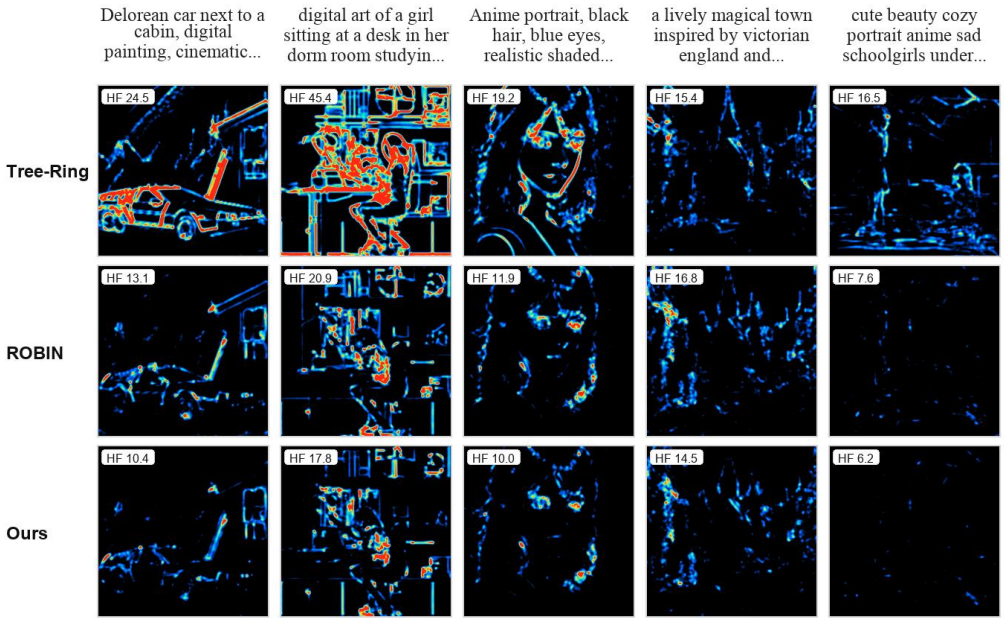


Figure 16: High-frequency error maps for watermarking methods.

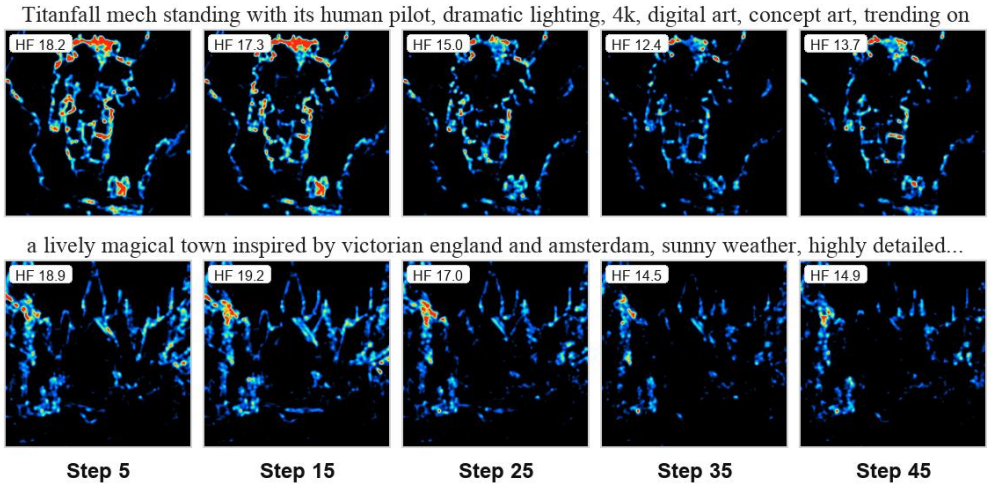


Figure 17: High-frequency error maps for cross-step transfer.

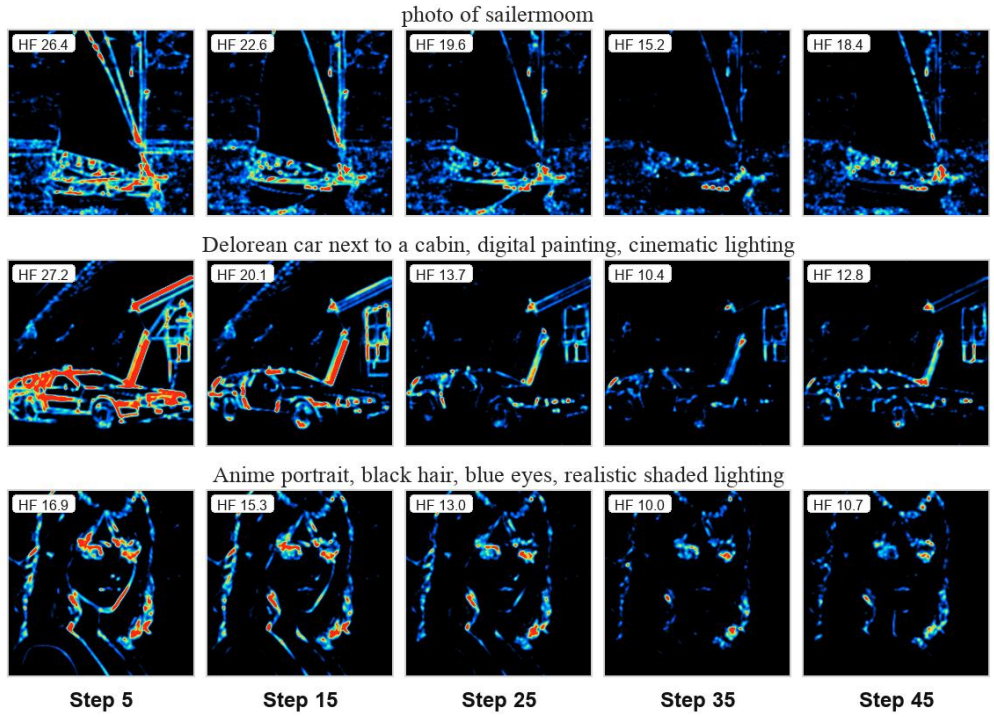


Figure 18: High-frequency error maps for cross-step transfer.

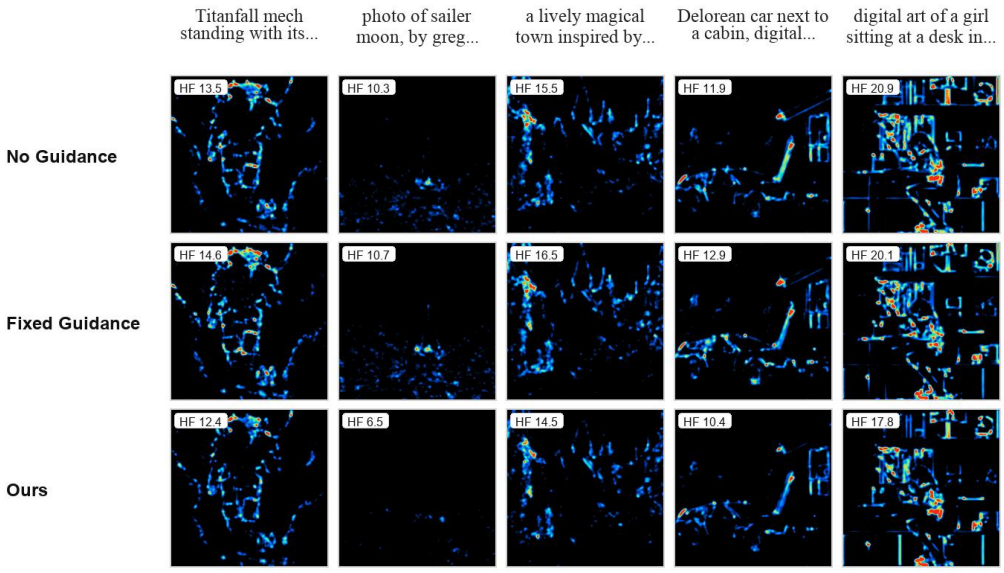


Figure 19: High-frequency error maps for different guidance strategies.

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and introduction state the paper's scope, method, and evaluation setting. The main claims concern trajectory-aware diffusion watermarking, content preservation, and robustness under image attacks, which are evaluated in Section 3 and Section 4.

   Guidelines:

   - The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
   - The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
   - The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
   - It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: Section 6 discusses limitations of the model-owner verification setting, including the need for a compatible inversion path, the lack of third-party verification, and open issues in payload capacity and key separation.

   Guidelines:

   - The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
   - The authors are encouraged to create a separate "Limitations" section in their paper.
   - The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
   - The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
   - The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
   - The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
   - If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
   - While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. **Theory assumptions and proofs**

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [N/A]

Justification: The paper does not present theoretical results such as theorems, lemmas, or formal guarantees. The equations and algorithms describe the proposed diffusion watermarking method and verification procedure rather than theoretical claims requiring proofs.

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. **Experimental result reproducibility**

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: The paper describes the model, datasets, baselines, evaluation metrics, attack settings, watermark construction, optimization procedure, and implementation details needed to reproduce the main experimental results.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general. releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
  (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
  (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
  (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).

(d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

   Answer: [No]

   Justification: The experiments use publicly available base models and datasets, and the paper reports the main implementation and evaluation details. They are correctly cited in the paper. We plan to release the code after publication.

   Guidelines:

   - The answer [N/A] means that paper does not include experiments requiring code.
   - Please see the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
   - While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
   - The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
   - The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
   - The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
   - At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
   - Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. **Experimental setting/details**

   Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

   Answer: [Yes]

   Justification: We provide experimental details in Section 4 and Appendix A such as the base model, datasets, baselines, metrics, and evaluation protocol.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
   - The full details can be provided either with the code, in appendix, or as supplemental material.

7. **Experiment statistical significance**

   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

   Answer: [Yes]

   Justification: We report variability for the main experimental results. The result tables report the mean over several independent runs, with standard deviation shown as a subscript. The runs vary the random seed used for watermark initialization and evaluation sampling.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. **Experiments compute resources**

   Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

   Answer: [Yes]

   Justification: Appendix A reports the GPUs used for experiments, and Table 5 reports generation and verification time per image/query.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
   - The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
   - The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. **Code of ethics**

   Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?

   Answer: [Yes]

   Justification: The research uses public models and datasets and focuses on watermarking generated images for provenance and verification. We have reviewed the NeurIPS Code of Ethics and do not identify deviations from it.

   Guidelines:

   - The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
   - If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
   - The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. **Broader impacts**

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: The paper discusses positive impacts for provenance, ownership verification, and responsible deployment of text-to-image models. We also discuss deployment limitations and risks related to verification scope, third-party verification, and large-scale attribution.

Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.
- If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. **Safeguards**

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: The work evaluates a watermarking method using existing public models and datasets. The paper does not release a new pretrained image generator, scraped dataset, or other high-risk asset.

Guidelines:

- The answer [N/A] means that the paper poses no such risks.
- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. **Licenses for existing assets**

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [Yes]

Justification: We cite the original sources for the base model, datasets, metrics, and baseline methods used in the experiments.

Guidelines:

- The answer [N/A] means that the paper does not use existing assets.
- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, `paperswithcode.com/datasets` has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. **New assets**

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [N/A]

Justification: The paper does not release new datasets or other standalone assets at submission time. The proposed method is described in the paper, and the experiments use existing public models and datasets.

Guidelines:

- The answer [N/A] means that the paper does not release new assets.
- Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
- The paper should discuss whether and how consent was obtained from people whose asset is used.
- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. **Crowdsourcing and research with human subjects**

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [N/A]

Justification: The paper does not involve crowdsourcing experiments or research with human subjects.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

    Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

    Answer: [N/A]

    Justification: The paper does not involve human subjects research, crowdsourcing, or collection of participant data.

    Guidelines:

    - The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
    - Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
    - We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
    - For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. **Declaration of LLM usage**

    Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

    Answer: [N/A]

    Justification: The core method development does not use LLMs as an important, original, or non-standard component. Any LLM use, if any, was limited to writing, editing, or formatting assistance and did not affect the methodology, experiments, or scientific claims.

    Guidelines:

    - The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
    - Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.